

Name: Sai Kiran Chowdary Unnam
Role: Senior AI/ML Engineer | Data Scientist
Email id: saikiranchowdaryunnam@gmail.com
Contact: +1 312-722-7883



Professional Summary

- Seasoned **AI/ML Engineer and Data Scientist** with over **9+ years of progressive experience** delivering scalable AI solutions across finance, healthcare, retail, telecom, and banking sectors. Proven expertise in architecting **end-to-end machine learning systems**, deploying models in production, and aligning AI initiatives with **regulatory standards** and business goals.
- Adept at working with **cloud-native ecosystems (AWS, Azure)**, advanced **deep learning frameworks (PyTorch, TensorFlow)**, and **MLOps best practices** to build explainable, high-performance AI applications in real-time environments.
- Designed and deployed real-time **fraud detection and risk modeling systems** in financial services using **Amazon Sage Maker, AWS Kinesis, and Kafka**, reducing operational risk and improving compliance.
- Led the development of **NLP models** using **Transformers and LLMs** to extract insights from unstructured financial and healthcare documents such as **SEC filings, analyst reports, and clinical notes**.
- Built and optimized **MLOps pipelines** using **MLflow, Kubeflow, and CI/CD tools (AWS CodePipeline, Azure DevOps)** to ensure reproducible, scalable, and governed ML deployments.
- Applied **deep learning techniques (CNNs, RNNs, and GNNs)** and **time-series forecasting models** for market behavior prediction and healthcare anomaly detection, leveraging **Tensor Flow and PyTorch**.
- Spearheaded **credit risk models and scorecards** at Credit One Bank, enhancing credit decision accuracy by 18% through robust modelling using **XGBoost, logistic regression, and random forest**.
- Engineered real-time **decision engines** and **microservices** using **AWS Lambda, Azure Functions, and Docker/Kubernetes**, enabling sub-second inference for loan approvals and fraud alerts.
- Implemented **automated data pipelines** with **AWS Glue, S3, Athena, and Azure Data Factory** to support high-throughput ML workflows with minimal latency and fault tolerance.
- Ensured **model explain ability and fairness** using **SHAP, LIME, and Amazon Clarify**, especially in high-stakes use cases like **credit scoring, medical diagnostics, and loan approvals**.
- Conducted continuous **A/B testing, drift monitoring, and automated retraining** using **Airflow and Sage Maker Pipelines**, ensuring sustained model performance and business alignment.
- Developed **interactive dashboards and visualizations** with **Tableau, Power BI, and AWS Quick Sight** to communicate AI outcomes effectively to senior leadership and business users.
- Maintained rigorous **compliance with FINRA, SEC, HIPAA, and GDPR** using built-in **cloud security and governance tools**, enabling safe and ethical AI deployment.
- Architected centralized **data lake platforms** on AWS and Azure to store and process structured and unstructured data, unlocking advanced analytics and ML capabilities across industries.
- Built **financial knowledge graphs** and **recommendation systems** to personalize client experiences and enhance asset allocation, leveraging **graph embedding's** and **semantic models**.
- Led efforts in **cost-optimization** of ML workloads through **Spot Instances, Savings Plans, and Cost Explorer**, reducing cloud expenditure while maintaining performance.
- Mentored junior engineers and data scientists across multiple organizations, fostering a collaborative culture of **continuous learning**, innovation, and excellence in AI/ML practices.
- Participated in large-scale **digital transformation initiatives**, modernizing legacy systems through **cloud migration strategies, MLOps automation, and data modernization programs**.
- Executed anomaly detection systems using **streaming analytics, unsupervised learning, and Azure Stream Analytics**, identifying behavioral outliers and fraud with high accuracy.
- Continuously evaluated the applicability of emerging technologies such as **Generative AI, Reinforcement Learning, and Multimodal ML** in capital markets, healthcare, and retail AI strategies.

Technical Skills

Programming Languages	Python, SQL
Machine Learning Frameworks	PyTorch, Tensor Flow, XGBoost, Scikit-learn
NLP & LLMs	Transformers, Large Language Models (LLMs), Semantic Models, Knowledge Graphs

Deep Learning Architectures	CNNs, RNNs, GNNs, Reinforcement Learning, Generative AI
Cloud Platforms	AWS (Sage Maker, Kinesis, Lambda, Glue, S3, Athena, Quick Sight), Azure (Functions, Data Factory, Stream Analytics)
MLOps & DevOps	MLflow, Kubeflow, CI/CD (AWS CodePipeline, Azure DevOps), Airflow, Sage Maker Pipelines
Big Data & Streaming	Kafka, AWS Kinesis, Spark, AWS Glue, Azure Stream Analytics
Containerization & Orchestration	Docker, Kubernetes
Model Explainability & Fairness	SHAP, LIME, Amazon Clarify
Data Storage & Processing	Data Lakes (AWS/Azure), SQL/NoSQL, Delta Lake
Time-Series & Forecasting	ARIMA, Prophet, LSTM, DeepAR
Visualization & BI Tools	Tableau, Power BI, AWS Quick Sight
Compliance & Security	FINRA, SEC, HIPAA, GDPR, Cloud Security Tools
Optimization & Cost Management	AWS Spot Instances, Savings Plans, Cost Explorer
Microservices & APIs	AWS Lambda, Azure Functions, REST APIs
Version Control & Collaboration	Git, GitHub/GitLab, Agile/Scrum

Educational Details

- Bachelor's in Information Technology at VNR VJIET - Hyderabad 2016

Certifications

- AWS Certified: Machine Learning Engineer – Associate
- Microsoft Certified: Azure Data Scientist Associate

Work Experience

Client: FINRA Rockville, MD
Role: Senior AI/ML Engineer

April 2024 - Present

Responsibilities:

- Design, develop, and deploy **scalable machine learning models** for real-time fraud detection, risk modelling, and portfolio optimization within financial systems.
- Collaborate with quant analysts, traders, and data scientists to align AI solutions with financial trading strategies and regulatory compliance standards.
- Implement **automated data pipelines** and feature engineering workflows using **AWS Glue, Lambda, and S3**, ensuring high availability and low latency.
- Leverage **Amazon SageMaker** for model training, tuning, deployment, and monitoring, while enforcing best practices in ML lifecycle management.
- Develop **NLP models** to extract insights from financial documents such as earnings transcripts, analyst reports, and SEC filings using **Transformers and LLMs**.
- Apply **deep learning techniques** (CNNs, RNNs, and GNNs) for time-series forecasting and market behavior prediction, using frameworks like **Tensor Flow and PyTorch**.
- Ensure model explain ability and fairness using tools like **SHAP, LIME**, and **Amazon Clarify**, especially in credit scoring and loan approval use cases.
- Maintain high model governance and auditability through **MLOps pipelines** with **MLflow, Kubeflow**, and **CI/CD** workflows integrated on **AWS CodePipeline**.
- Collaborate with DevOps and Data Engineering teams to manage **containerized ML deployments** using **Docker, Kubernetes**, and **EKS on AWS**.
- Architect and manage **real-time inference systems** leveraging **AWS Kinesis, Kafka**, and **API Gateway**, ensuring sub-second decision-making capabilities.
- Monitor model drift and data quality degradation through continuous **A/B testing** and **automated retraining workflows** using **Airflow** and **Sage Maker Pipelines**.
- Build financial knowledge graphs and recommendation engines to enhance client personalization and asset allocation advisory.

- Lead code and model reviews to uphold best coding practices and ensure reproducibility, scalability, and performance of ML workloads.
- Conduct **security and compliance assessments** for AI systems in line with **FINRA, SEC, and GDPR** requirements using AWS-native tools.
- Stay up-to-date with innovations in **Generative AI, Reinforcement Learning, and Multimodal ML**, assessing their applicability in capital markets.
- Drive cost-optimization of AI workloads by leveraging **Spot Instances, Compute Savings Plans, and AWS Cost Explorer** analytics.
- Mentor junior engineers and data scientists, fostering a culture of innovation, collaboration, and continuous learning within the AI/ML division.
- Present AI-driven financial insights and ROI evaluations to senior leadership, enabling data-informed strategic decisions.

Client: UHG Marietta, GA

Jan 2022 – Mar 2024

Role: AI/ML Engineer

Responsibilities:

- Developed and deployed **machine learning models** for predictive analytics in patient care, improving diagnosis accuracy and operational efficiency in healthcare workflows.
- Designed and maintained **end-to-end ML pipelines** using **AWS Sage Maker**, ensuring scalable and secure deployment of real-time models in production.
- Leveraged **AWS services** including **EC2, Lambda, S3, Athena, and Glue** for managing large-scale healthcare data and building ML inference APIs.
- Applied **NLP techniques** on unstructured clinical notes and medical transcripts to extract key insights, enhancing patient care outcomes and billing efficiency.
- Implemented **deep learning models** using **Tensor Flow** and **PyTorch** for disease prediction and medical image classification, improving diagnostic support systems.
- Integrated **CI/CD pipelines** using **AWS CodePipeline** and **Code Build** to automate testing and deployment of ML workflows, reducing release cycles.
- Developed **data pre-processing** frameworks to clean, normalize, and transform high-dimensional patient data, supporting accurate and unbiased model training.
- Collaborated with cross-functional teams including data scientists, cloud architects, and healthcare SMEs to align ML solutions with clinical requirements.
- Led efforts to **optimize model performance** through hyper parameter tuning, feature engineering, and validation strategies using **Sklearn** and **XGBoost**.
- Ensured full **HIPAA compliance** and data governance standards in model development to safeguard sensitive healthcare data and meet regulatory requirements.
- Created **interactive dashboards** and visualizations using **AWS Quick Sight** and **Tableau** to communicate model outcomes to business stakeholders.
- Conducted **model interpretability** analysis using SHAP and LIME techniques, enabling transparent decision-making and trust in AI-assisted diagnostics.
- Utilized **Docker and Kubernetes** for containerization and orchestration of ML services, enhancing portability and system resilience in healthcare apps.
- Performed **time-series analysis** and anomaly detection on patient vitals and lab data to proactively detect deteriorating conditions.
- Maintained **version control of ML models** and datasets using **DVC** and **Git**, ensuring reproducibility and traceability of model iterations.
- Implemented **data lake architecture** on AWS for centralized storage of structured and unstructured healthcare data, enabling advanced analytics and ML integration.
- Participated in **market-aligned AI initiatives**, benchmarking healthcare AI solutions against industry leaders to ensure competitive innovation and value delivery.
- Delivered comprehensive documentation, training, and knowledge transfer to internal stakeholders, enabling smooth transition and model lifecycle continuity.

Client: First citizen bank, Charlotte, NC

May 2018 – Dec 2021

Role: Senior Data Scientist

Responsibilities:

- Spearheaded the end-to-end development of predictive models for credit risk assessment, fraud detection, and customer segmentation, aligning closely with regulatory compliance requirements in the banking sector.

- Utilized **Azure Machine Learning**, **Azure Data Factory**, and **Azure Synapse Analytics** to orchestrate scalable ML pipelines, ensuring high availability and efficiency across cloud environments.
- Collaborated with credit risk teams to design statistically robust scorecards using **logistic regression**, **XGBoost**, and **random forest**, improving default prediction accuracy by 18%.
- Developed real-time decision engines using **Azure Functions** and **Event Hubs** to deliver instant credit approval decisions, optimizing loan disbursement turnaround time.
- Applied **Natural Language Processing (NLP)** on customer support transcripts to identify churn signals and customer satisfaction themes, aiding retention strategy.
- Integrated **Power BI** dashboards with machine learning outputs to provide actionable insights for business stakeholders, driving data-informed decisions.
- Designed customer lifetime value (CLV) models using **Python**, **Azure Data bricks**, and advanced time series forecasting methods to improve long-term profitability predictions.
- Partnered with DevOps and data engineering teams to deploy models into production using **MLOps best practices**, leveraging Azure DevOps CI/CD pipelines.
- Ensured all model development adhered to **Fair Lending regulations**, explainability standards like SHAP/LIME, and model governance policies across banking operations.
- Led advanced experimentation and A/B testing frameworks to evaluate campaign performance and personalize product offerings for different credit profiles.
- Conducted exploratory data analysis (EDA) and feature engineering on structured and unstructured data using **SQL**, **Pandas**, and **PySpark**, enhancing model precision.
- Executed anomaly detection systems to flag unusual spending behavior using unsupervised learning and streaming analytics tools on **Azure Stream Analytics**.
- Mentored junior data scientists and analysts on best practices in data science, model validation, and cloud-native ML development workflows.
- Optimized customer acquisition funnels through **uplift modelling** and conversion rate analysis, integrating campaign data with Azure Data Lake for scalable computation.
- Collaborated with cross-functional teams including finance, legal, and IT to ensure alignment of data science initiatives with organizational KPIs and regulatory frameworks.
- Implemented credit line increase (CLI) models using behavioural scorecards, improving credit utilization while minimizing exposure to high-risk segments.
- Built interactive model monitoring systems with **Azure Monitor**, **Application Insights**, and custom telemetry, reducing model downtime and increasing transparency.
- Participated in enterprise-level digital transformation initiatives, advocating for cloud migration strategies and championing **Azure-based data modernization** programs.

Client: Lulu lemon, Dallas, TX

Sep 2017 – April 2018

Role: Data Scientist

Responsibilities:

- Developed predictive models for customer segmentation using **Python**, **SQL**, and **Azure Machine Learning**, enabling more personalized marketing and increasing customer engagement by over 20%.
- Collaborated with merchandising and supply chain teams to design demand forecasting models using **Azure Data bricks** and **Time Series Analysis**, resulting in improved inventory turnover rates.
- Engineered data pipelines on **Azure Data Factory** to automate ingestion, transformation, and storage of structured and unstructured retail data, boosting data availability and reliability across departments.
- Led the implementation of **recommendation systems** using **Spark MLlib** and **Azure HDInsight**, improving cross-sell and upsell performance in online retail platforms.
- Created dashboards and visualization reports using **Power BI** integrated with **Azure Synapse Analytics**, driving actionable insights for executives on sales trends and customer behavior.
- Conducted **A/B testing and uplift modelling** to evaluate campaign effectiveness, applying statistical rigor and reducing marketing costs by 15%.
- Utilized **Azure Cognitive Services** for sentiment analysis on customer reviews, informing product development and improving customer satisfaction scores.
- Designed classification models to predict customer churn using **Scikit-learn**, **Azure ML Studio**, and advanced feature engineering techniques, achieving 92% accuracy.
- Partnered with IT to ensure data governance and compliance standards across cloud data environments, aligning with corporate risk policies and enhancing data security.

- Applied **Natural Language Processing (NLP)** to extract themes from customer feedback, aiding in trend spotting and proactive service improvements.
- Collaborated with pricing analysts to build dynamic pricing models using **R**, **Azure SQL Database**, and market elasticity curves, optimizing profit margins.
- Conducted clustering and segmentation analysis using **K-means** and **Hierarchical Clustering** techniques, supporting tailored marketing efforts across customer cohorts.
- Mentored junior analysts on **data wrangling**, **feature selection**, and **model validation** using industry-standard tools, contributing to team capability growth.
- Integrated third-party retail market data through **Azure Data Lake Storage** for competitive benchmarking, enriching decision-making on product assortment.
- Deployed end-to-end ML solutions from model development to monitoring using **Azure DevOps**, **MLflow**, and **CI/CD pipelines**, ensuring production readiness.
- Interacted closely with cross-functional stakeholders to translate business challenges into data-driven solutions, ensuring alignment between analytics and organizational goals.

Client: BSNL, Hyderabad India

Feb 2016 – Jun 2017

Role: Data Engineer

Responsibilities:

- Designed and developed **robust ETL** pipelines to automate ingestion and transformation of large-scale telecom data from diverse sources, enhancing real-time decision-making.
- Collaborated with cross-functional teams to architect and implement scalable data warehouse solutions using Amazon **Redshift** and **SQL**-based schema design tailored to telecom analytics needs.
- Developed and deployed batch and streaming data processing workflows using Apache Spark and Kafka, enabling near real-time customer usage insights and call quality metrics.
- Maintained and optimized large volumes of telecom data within Hadoop Distributed File System (**HDFS**) ensuring efficient retrieval, fault-tolerance, and cost-effective storage.
- Engineered and maintained data integration frameworks for provisioning subscriber metadata, call records, and network logs, supporting internal business intelligence platforms.
- Implemented data quality checks, validation logic, and monitoring frameworks to ensure telecom regulatory compliance and accuracy of critical usage reports.
- Utilized Python and Scala scripts to automate data manipulation tasks and deliver reproducible pipelines, significantly reducing manual intervention.
- Worked closely with analytics and product teams to build reliable data models and marts, tailored to VoIP performance metrics, churn prediction, and customer behavior analysis.
- Established **CI/CD** workflows for data jobs using Jenkins, ensuring seamless deployment and version control of data pipelines in production environments.
- Collaborated in the adoption of cloud-native solutions (**AWS S3, EMR, Lambda**) to migrate legacy workloads, increasing performance, scalability, and operational agility.
- Applied data partitioning, indexing, and optimization strategies to improve the query performance of time-series-based telecom datasets by over 40%.
- Supported data security by implementing role-based access control (**RBAC**), encryption standards, and data governance policies within engineering frameworks.
- Worked with business intelligence tools like Tableau and Looker to enable visualization of customer call logs, VoIP trends, and infrastructure **KPIs**.
- Contributed to the development of machine learning pipelines by preparing feature datasets for network anomaly detection and user segmentation models.
- Regularly participated in code reviews, sprint planning, and architectural discussions, driving improvements in engineering practices and telecom-specific data architecture standards.